

Escambia County — Coverage, Cost & Margins

INTERNAL / PARTNER COPY — contains OCWS margin data. Do not distribute to customer.

A modeled county-wide gunshot-detection rollout: coverage, program cost, our pricing, and 5-year value vs ShotSpotter. Three sensor tiers on one command picture. Figures are a planning model, not a committed bid (four numbers need live quotes — see last page).

~1,312 ENDPOINTS, COUNTY-WIDE	~\$1.46M CUSTOMER 5-YR (OWNED)	~\$452K OCWS 5-YR GROSS (31%)	5-12x CHEAPER THAN SHOTSPOTTER
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01 How it works

Cheap microphone **leaves** blanket each building so every space is heard. When a shot fires, the nodes that hear it each report over LoRa to one **gateway**, which fuses them into *one* located alert to the C2 in ~1-2 seconds — one banner, not one per sensor. Every patrol cruiser is also a mobile gateway, so coverage travels with the deputies. Location precision rises with deployment:

- **Ships first:** which nodes heard it -> the shot is in that area. Already beats a 911 call.
- **Phase 2:** a region / uncertainty ellipse from combined loudness.
- **Phase 3:** a precise GPS point via time-difference-of-arrival across GPS-timed nodes.

02 County-wide coverage — three tiers

Tier	Sites / assets	Endpoints	Materials
A — County facilities (courthouse flagship, admin, libraries, ECSO)	35	301	\$48,230
B — Schools (7 high, 8 middle, 34 elementary, 6 alt) — grant-funded	55	751	\$117,890
C — Mobile (~250 cruisers + 10 drones) — coverage travels	260	260	\$73,100
County-wide		~1,312	~\$239,220

Materials only. Per-building node counts are planning estimates; the Crow's Eye LiDAR survey produces exact counts + the coverage map that becomes the acceptance document.

03 Pricing — three revenue lines

Hardware is priced thin to win the deal; the **Crow's Eye survey** and the **recurring service** are the margin engines. Install is outsourced and marked up — not eaten out of hardware.

Revenue line	Customer pays	OCWS gross
Hardware (capex, owned by customer; priced 1.25x cost)	~\$341,000	~\$68,000
Install (outsourced integrator + 1.3x markup)	~\$268,000	~\$62,000
Crow's Eye survey (~90 buildings; ~73% margin)	~\$99,000	~\$72,000
One-time subtotal	~\$708,000	~\$202,000
Annual service — C2 + monitoring + maintenance (x5 yr)	~\$750,000	~\$250,000
5-YEAR TOTAL	~\$1,458,000	~\$452,000

Blended 5-yr margin ~31%. The annual service compounds and the whole model repeats per county; contract-manufacturing improves margin at scale. Battery-replacement cadence is the #1 maintenance-cost lever (truck rolls dominate) — a LiFePO4 / trickle-power design decision.

04 5-year value vs ShotSpotter (the billing argument)

	Coverage	5-year cost	Owned?
ShotSpotter (20-50 mi² urban hot-zones)	outdoor gunshot only	~\$7M - \$18M	No - rented
Sentinel (county-wide)	interiors + county + mobile	~\$1.5M	Yes - owned

ShotSpotter is \$65-90K per mi²/yr, always rented, only the zones a city pays for (NYC \$85,097/mi²/yr, verified). Sentinel is owned, county-wide, and auditable (public corpus, re-runnable harness, 97.8% recall). The competitor mi² figures are secondary except NYC's — pitch the structural gulf, not a precise quote.

05 Acquisition — contract close to installed

Stage	Hand-build (pilot / Phase 1)	CM (county scale)
Procure components	2-6 wk	4-12 wk
Assemble + QA	~2-4 wk (~300 units)	4-8 wk (8-12 first build)
Crow's Eye survey + schedule	3-6 wk	4-8 wk (90 buildings)
Install (outsourced crews)	2-4 wk	6-16+ wk
Net close -> installed	~2-4 months	~4-7 months

Stages overlap. Hand-build the pilot + Phase 1; move to a contract manufacturer at the 1,000+ county run (early revenue funds the tooling). Crew count is the install-schedule lever. Supply: the 2021-23 shortage is over; verify Pi / SX1262 / MEMS mic / cells live before a volume order.

Four numbers to close before this is a bid: (1) live volume component prices/lead-times; (2) real per-vehicle upfit labor; (3) contract-manufacturer per-unit quote; (4) ECSO standing fleet count (scales the mobile tier). Full detail: docs/SENTINEL-ESCAMBIA-CASE-STUDY.md and docs/SENTINEL-PRICING-MODEL.md.